

Macrosystems EDDIE: Time Series Modeling and Prediction of Environmental Data

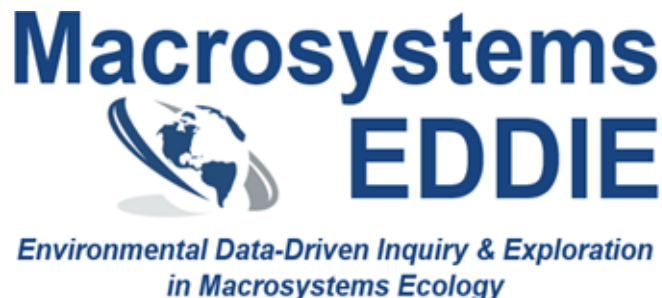
Lofton, M.E., M.R. Hipsey, K. Kurucz, and C.C. Carey. 8 August 2025.

Macrosystems EDDIE: Time Series Modeling and Prediction of Environmental Data.

Macrosystems EDDIE Module 11, Version 1.

https://serc.carleton.edu/dev/eddie/teaching_materials/modules/module11.html

Module development supported by NSF EF 2318861, DEB 2213550, OISE 2330211



Swan-Canning Estuary Virtual Observatory



neon
Operated by Battelle

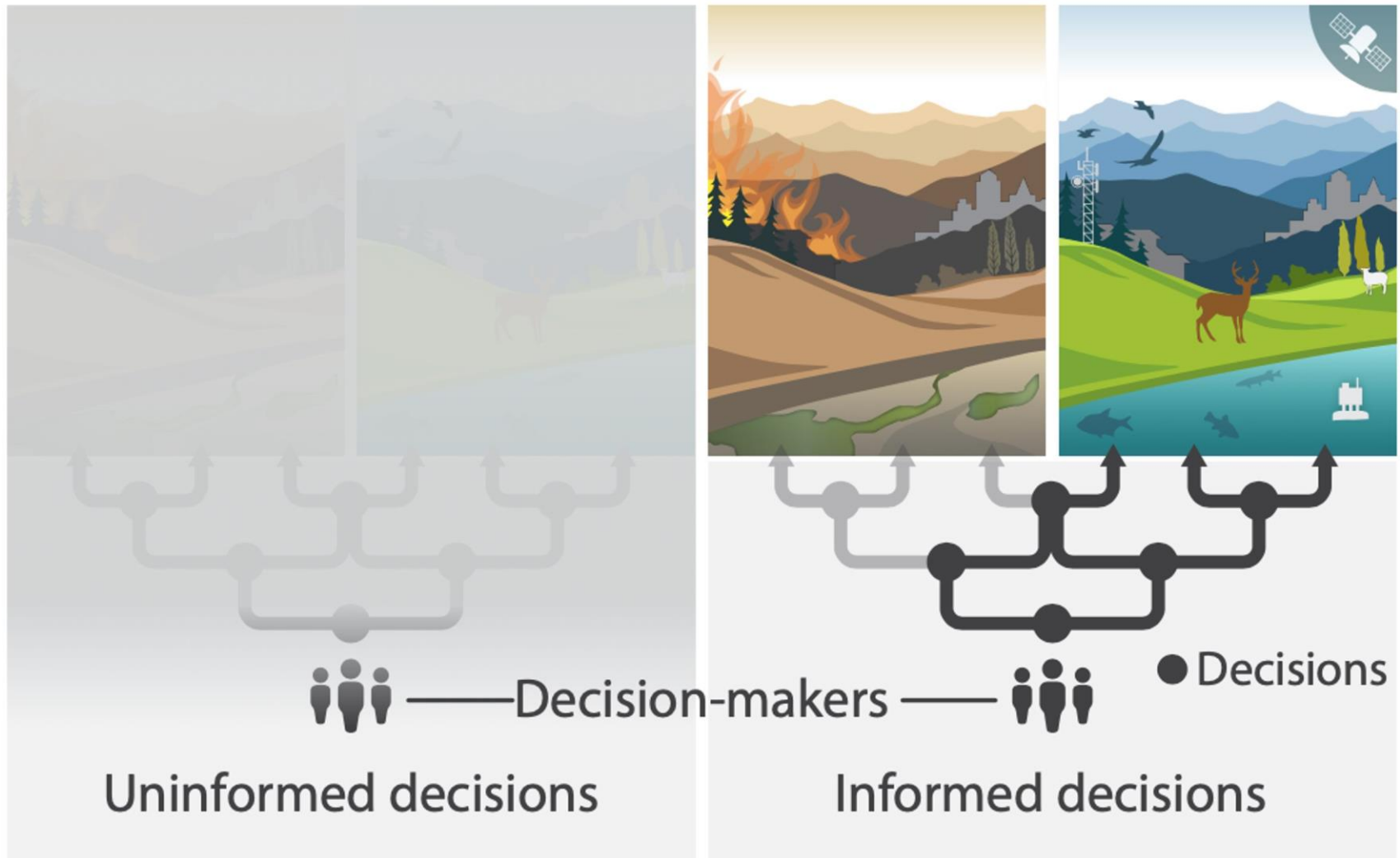
Overview of today

- Introduce concepts related to time series modeling and environmental data case studies
- **Activity A:** Visualize data from a selected environmental data case study, and fit and assess a time series model
- **Activity B:** Choose a new environmental dataset or upload your own dataset and fit and assess a time series model
- **Activity C:** Fit additional time series models to your environmental dataset from Activity B and compare performance across models

Our focal question for today:

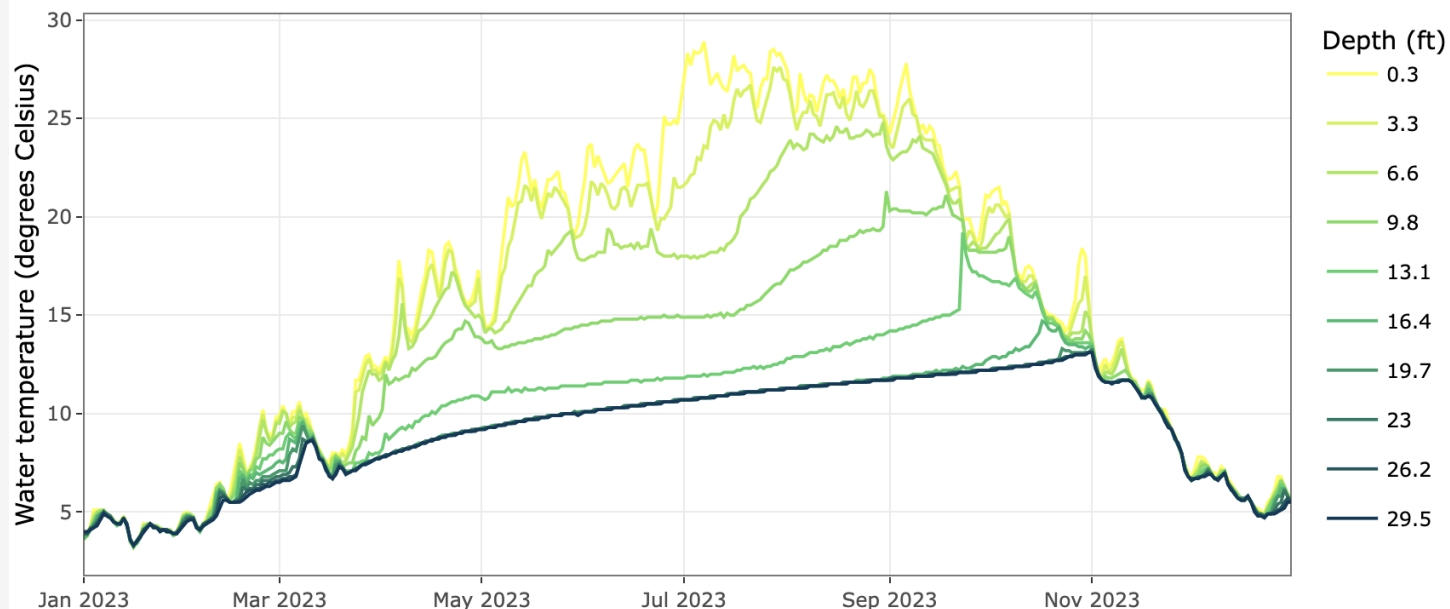
How can we use time series models to understand and predict ecosystem conditions?

Importance and urgency of environmental prediction in a changing world



High-frequency environmental data can be used to develop predictive models

- Measurements taken many times per day or week (for example, every 10 minutes or every day)
- Allows managers and scientists to see patterns that were not observable using low-frequency data (for example, once a week or once a month)



Time series models are often applied to analyze and make predictions using environmental data



Salmon population size



Airborne pollen concentration



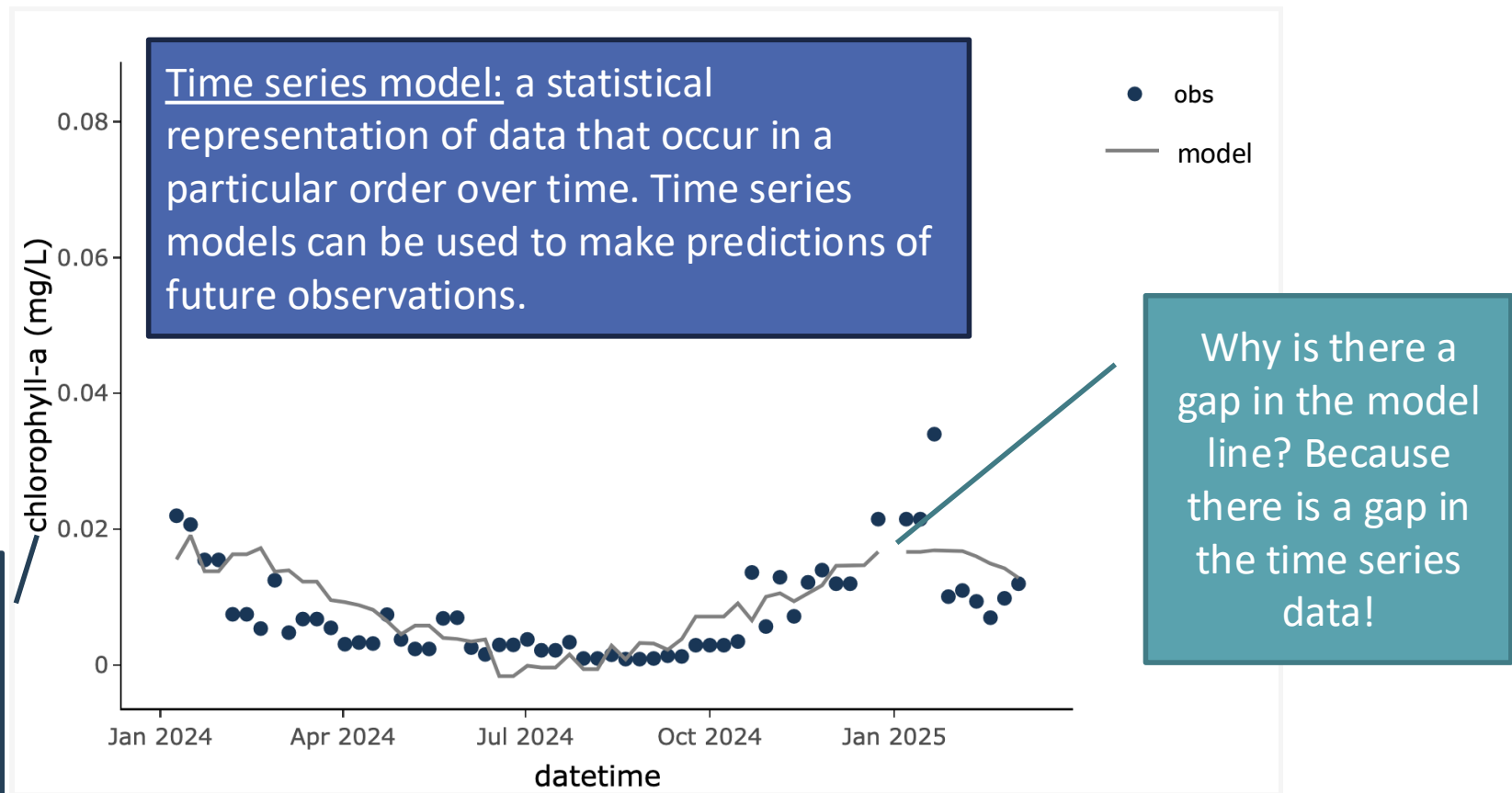
Forest pest outbreaks



Harmful algal blooms

Today...

We will explore **time series modeling** and out-of-sample prediction of standardized environmental datasets using several different models.

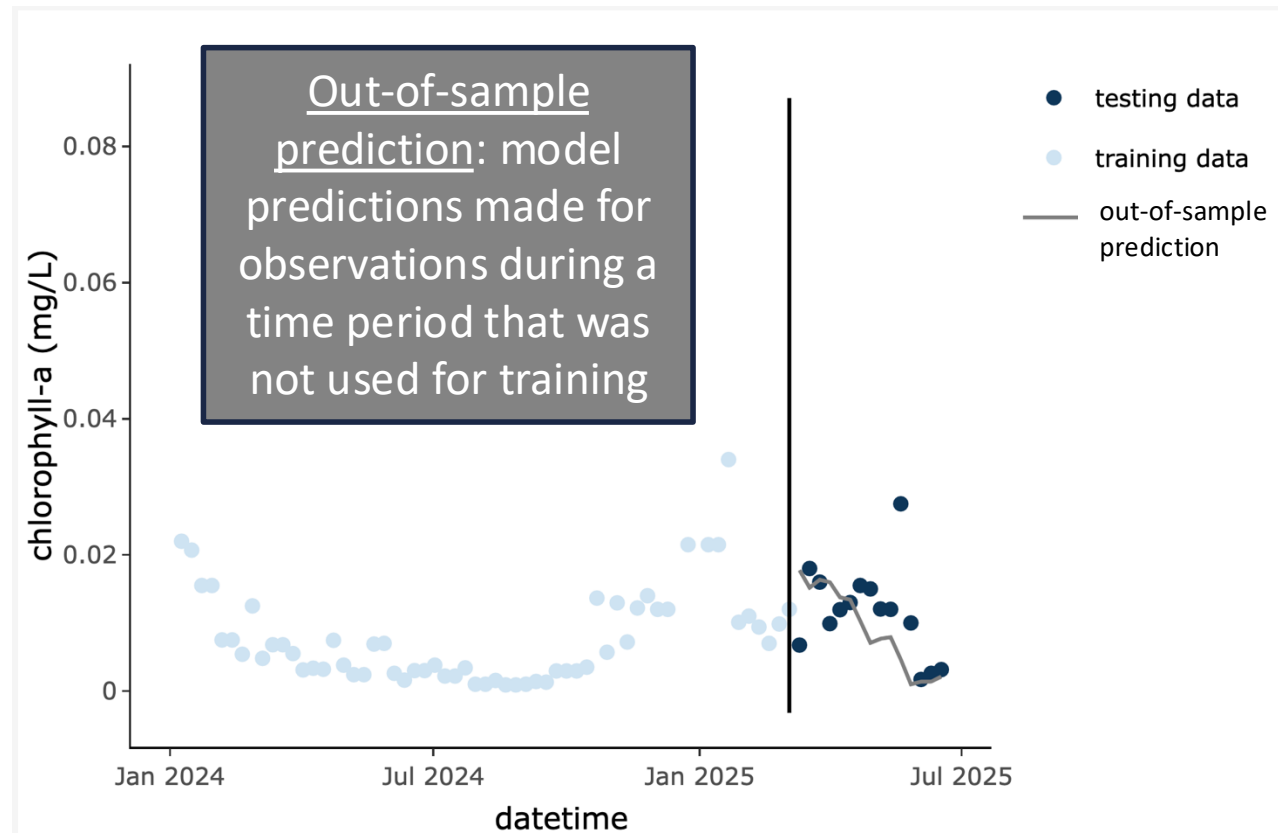


Today...

We will explore time series modeling and **out-of-sample prediction** of standardized environmental datasets using several different models.

Training data:
observations used to
fit a model

Testing data:
observations held “in
reserve” to evaluate a
model after it has
been developed



Today...

We will explore time series modeling and out-of-sample prediction of **standardized environmental datasets** using several different models.



Good! Dates are formatted consistently and there are only numeric values in the temperature column.

date	temp_degC
2021-08-03	25.6
2021-08-04	25.8
2021-08-05	



Bad! Dates are not consistently formatted and there is a mix of numeric and character entries in the temperature column. A computer won't know how to handle this without extra work on your part! 😞

Date	Temperature
8/3/21	25.6
8/4/21	25.8
Thursday, August 5, 2021	sensor broke

Today...

We will explore time series modeling and out-of-sample prediction of standardized environmental datasets using **several different models**.

- Researchers use different types of models because they provide complementary predictions: some may be more accurate than others, and provide unique information about an ecosystem
- Commonly-used ecological time series models include statistical time series models, machine learning models, and baseline models

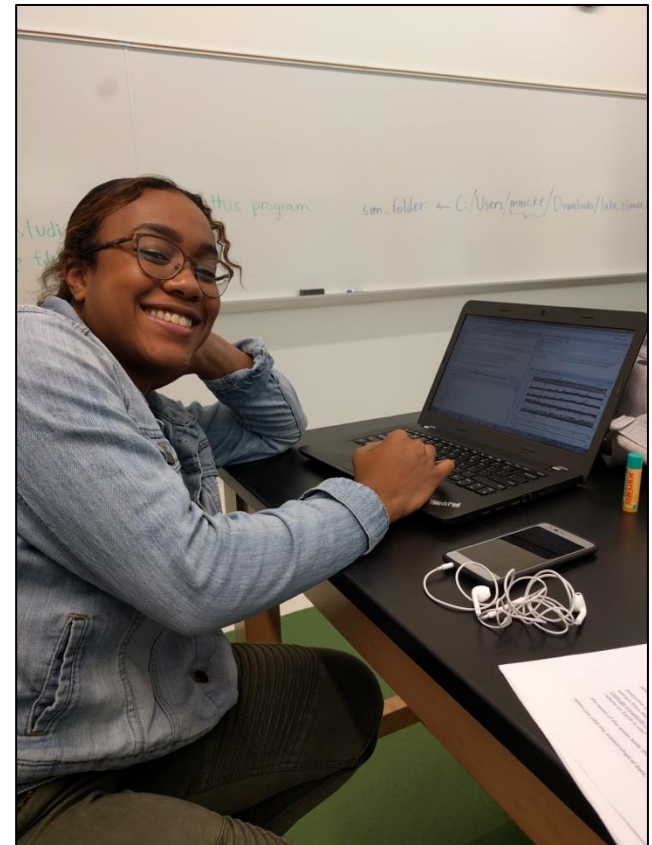


Photo credit: Megan Kelly, Arrupe College of Loyola University Chicago

Today...

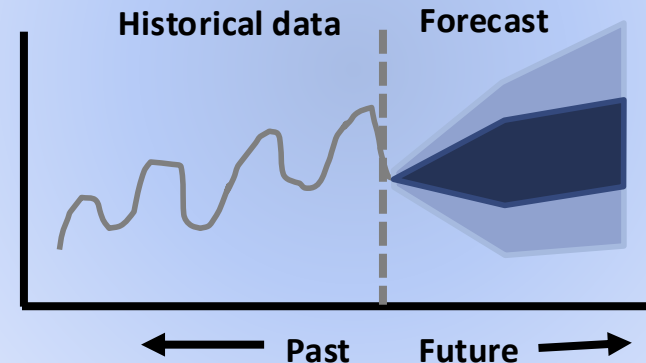
We will explore time series modeling and out-of-sample prediction of standardized environmental datasets using **several different models**.

Models we will use:

1. ARIMA
(AutoRegressive
Integrated Moving
Average)

AutoRegressive

ARIMA models use past values of a variable to predict future values



Integrated

The target variable may be **differenced**, meaning that each number is subtracted from the one after it

Moving Average

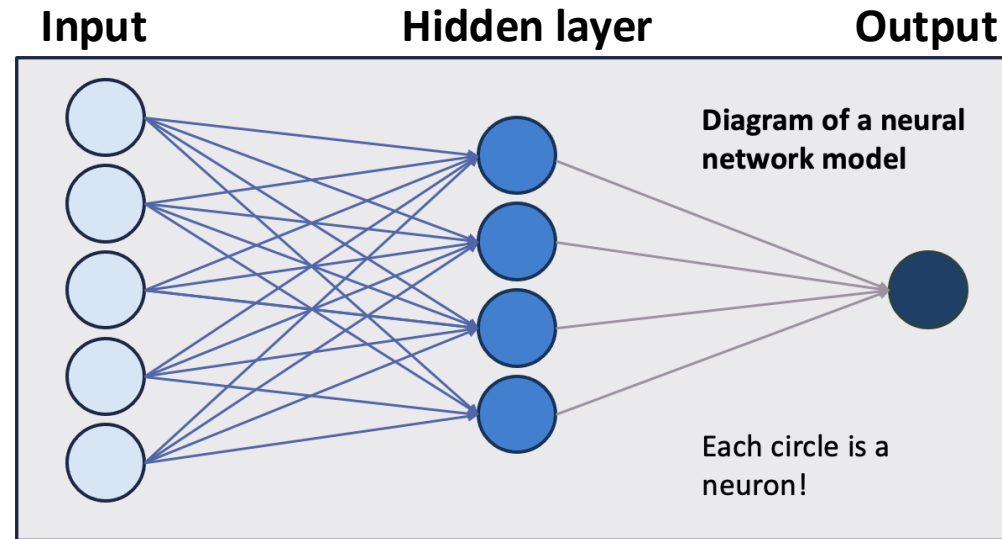
ARIMA models use past model errors to help make predictions of future values

Today...

We will explore time series modeling and out-of-sample prediction of standardized environmental datasets using **several different models**.

Models we will use:

1. ARIMA
2. NNETAR
(Autoregressive Neural Network)



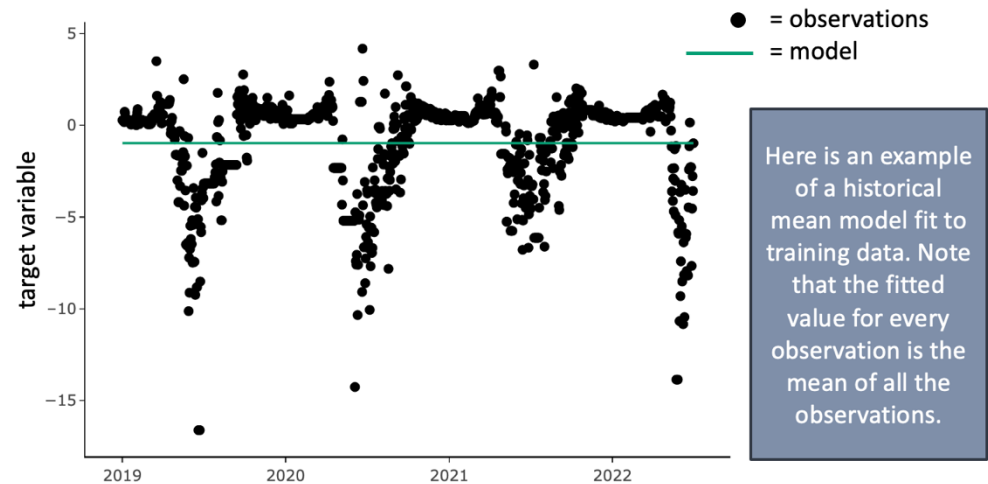
A **neural network** is a model whose structure is designed to mimic the structure of the human brain and nervous system. They are good at capturing complex, nonlinear relationships in data.

Today...

We will explore time series modeling and out-of-sample prediction of standardized environmental datasets using **several different models**.

Models we will use:

1. ARIMA
2. NNETAR
3. Historical mean



A **historical mean model** uses the mean of the previous observations as the prediction for the next timestep.

Historical mean models can be useful baseline models against which to compare more complex models.

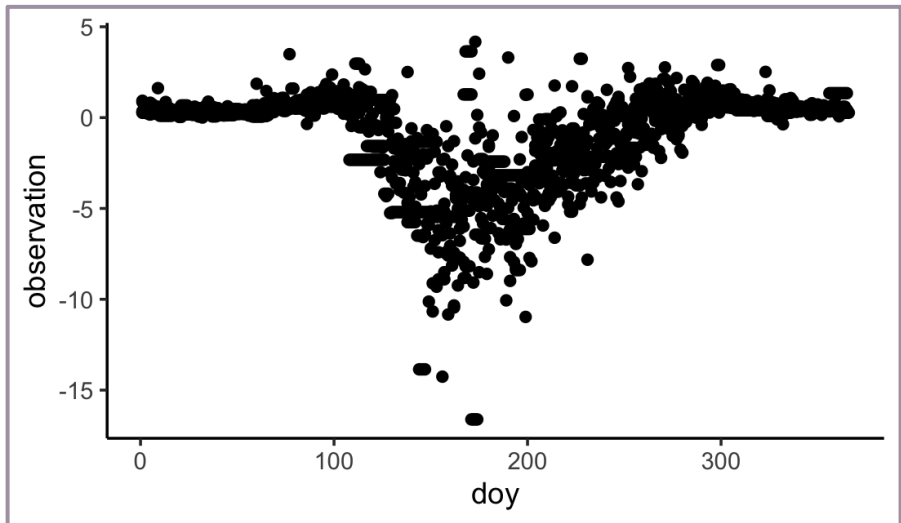
Today...

We will explore time series modeling and out-of-sample prediction of standardized environmental datasets using **several different models**.

Models we will use:

1. ARIMA
2. NNETAR
3. Historical mean
4. DOY (day-of-year)

Example figure of net ecosystem exchange (NEE) observations vs. day-of-year (day).

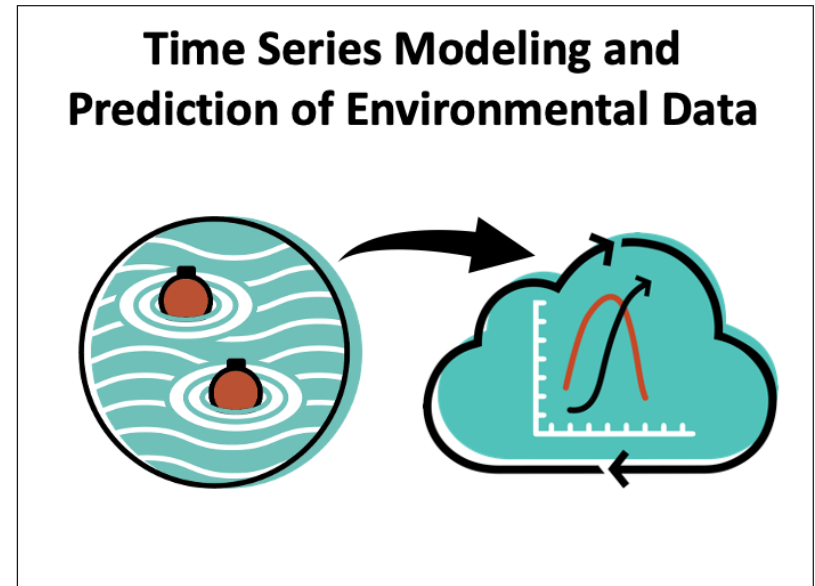


A **day-of-year model** uses the Julian day of the year to predict the target variable.

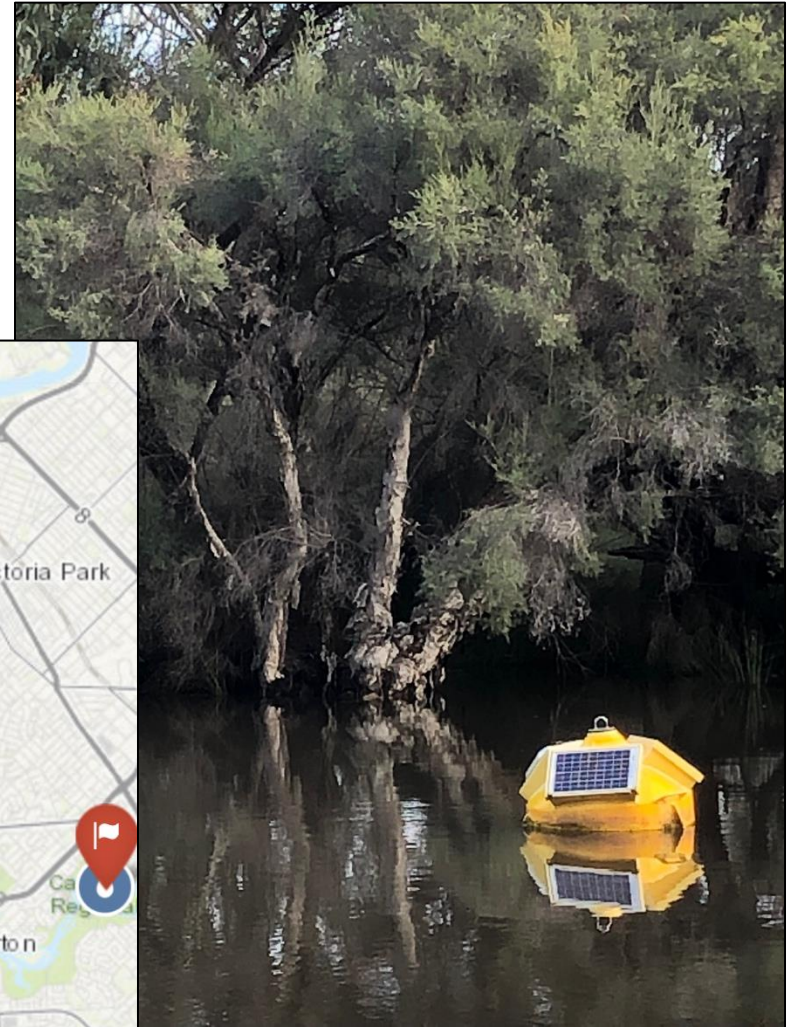
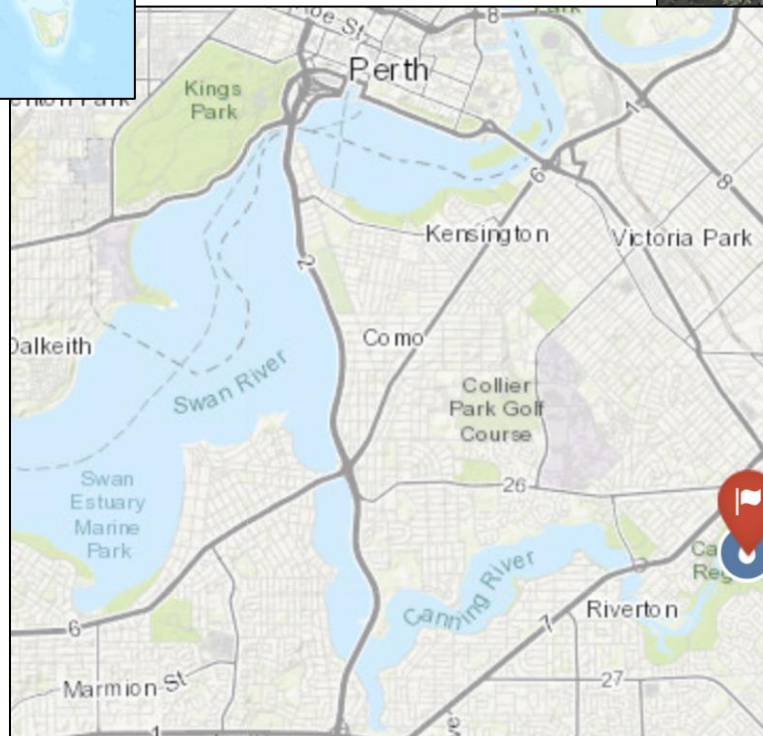
The Julian day ranges from 1 for Jan. 1 to 365 for Dec. 31 (or 366 if it is a leap year!).

Learning objectives of today's module:

1. Explore relationships between ecological variables from environmental case studies.
2. Understand the structure of four time series models (including machine learning models) that are commonly applied in environmental science.
3. Fit time series models using environmental data and assess the importance of ecological driver variables in making model predictions.
4. Process environmental datasets into standardized formats suitable for training and assessing time series models.
5. Compare multiple time series models to assess their performance on out-of-sample predictions.

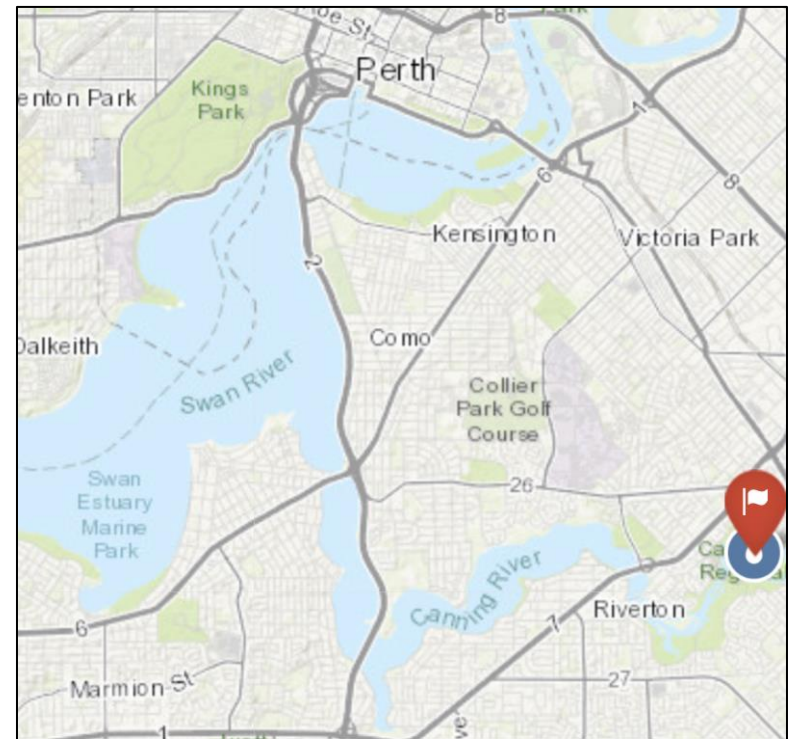


Aquatic case study: Canning River, WA, AU



Aquatic case study: Canning River, WA, AU

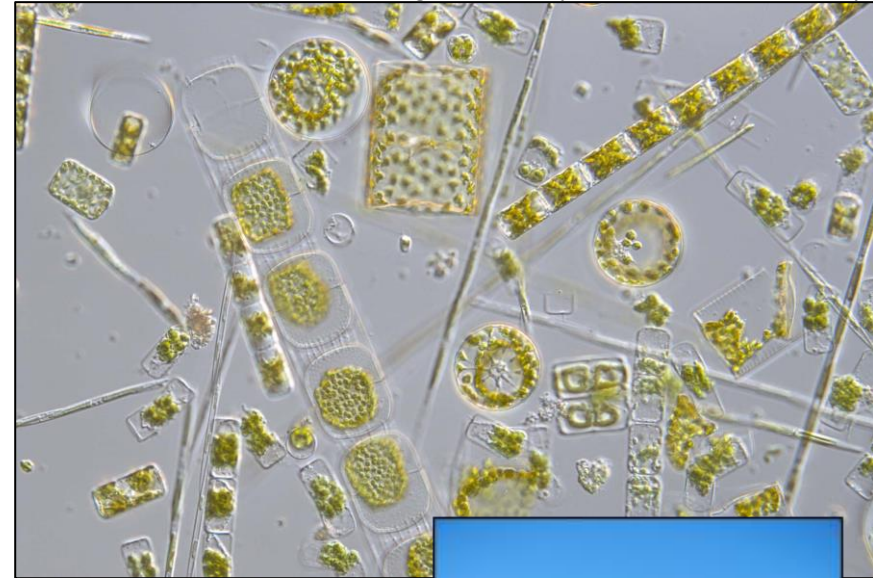
- Part of the Swan-Canning Estuary
- Urban estuary system that runs through Perth, the capital of Western Australia
- Freshwater flushing from terrestrial catchments and salt water intrusion from the Indian Ocean leads to complex hydrology and biogeochemistry
- Increasing nutrient loading from human activity, resulting in water quality impacts for this drinking water supply



Target variable: phytoplankton (chlorophyll-a)

- **Phytoplankton** are key primary producers in aquatic ecosystems
- **Chlorophyll-a** is a pigment used in photosynthesis and a proxy for **phytoplankton** biomass
- However, proliferation of **phytoplankton** can lead to problematic **blooms**
- **Blooms** can release toxins, cause taste and odor problems, and form unsightly scums

Phytoplankton under a microscope
Image source: Plymouth Marine Laboratory



Chlorophyll-a is measured using high-frequency sensors which can be deployed on buoys



Photo credit: Jonathan Thom

Management implications of chlorophyll-a

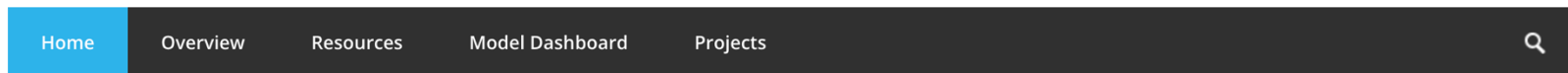
- The Swan-Canning estuary has historically experienced harmful blooms of different phytoplankton species
- This can lead to fish kills, fisheries warnings, recreational area closures, and other impairment
- Prediction of high chlorophyll-a levels may indicate an impending bloom



Data source: Swan-Canning Estuary Virtual Observatory (SCEVO)



Swan-Canning Estuary Virtual Observatory



Estuary Science

The SCEVO initiative aims to further our understanding and appreciation of the Swan-Canning Estuary in Perth, Western Australia



Model Simulations

This site displays a variety of historical, recent and forecast simulations of the Swan-Canning Estuary, based on 3-D simulations of hydrodynamic-biogeochemical models.



Information Portal

SCEVO serves as a central information portal to find out more about the Swan-Canning system.

The Swan-Canning Estuary lies at the heart of Perth, Western Australia

This site is an portal to help inform the community how it works and what it is doing

<https://swan.science.uwa.edu.au/>

Terrestrial case study: Bartlett Experimental Forest, NH, USA



NEON.D01.BART.DP1.00033 - NetCam SC IR - Sun Jul 27 2025 00:00:05 UTC
Camera Temperature: 47.0
Exposure: 1407

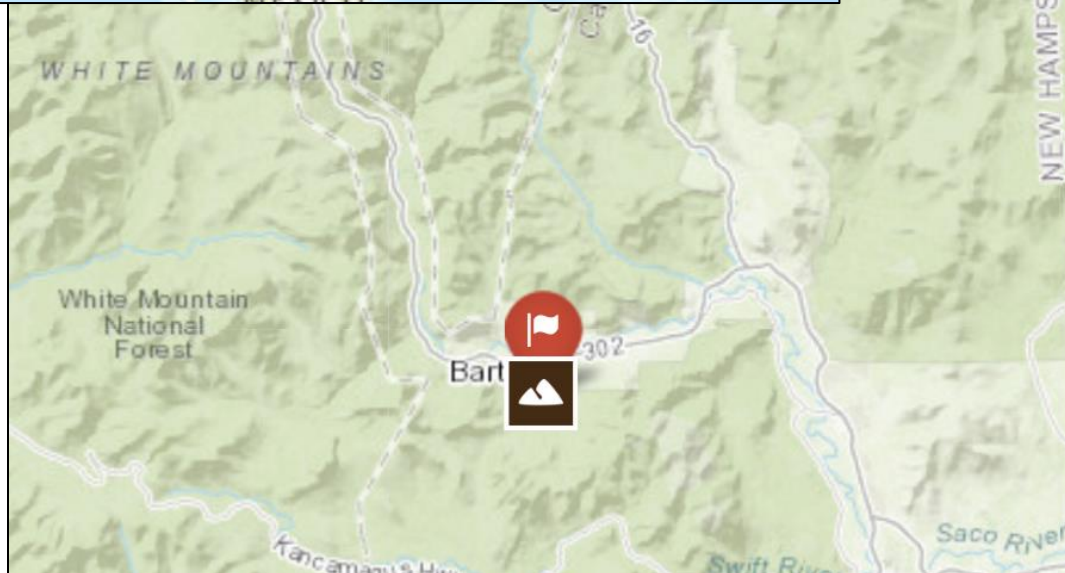
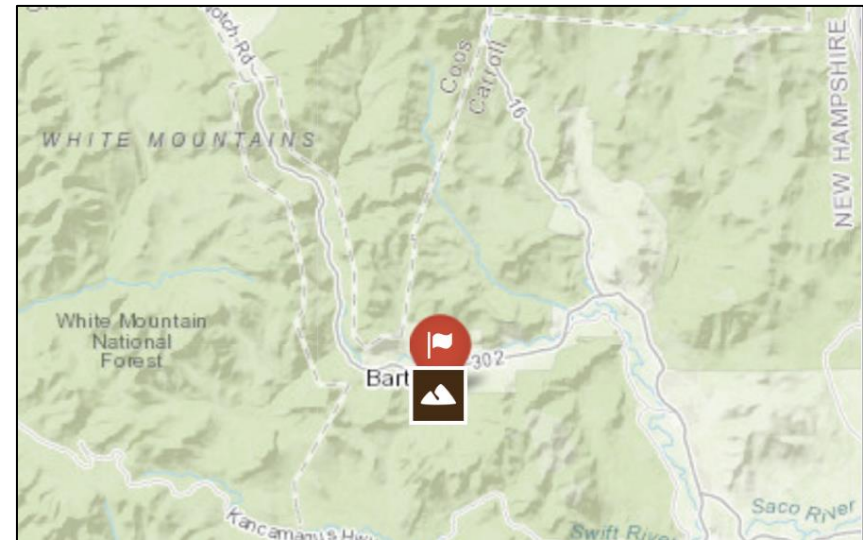


Image credit: U.S. National Ecological Observatory Network (NEON)

Terrestrial case study: Bartlett Experimental Forest, NH, USA

- Old-growth northern deciduous hardwood forest (beech, maple)
- Located within the White Mountain National Forest
- Historically logged
- Research monitoring station established in 1931
- Managed by the U.S. Department of Agriculture Forest Service and monitored as part of NEON



neon
Operated by Battelle

Data source: U.S. National Ecological Observatory Network (NEON)

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NEON data inform
the nation's safety,
security and well-
being.

The National Ecological Observatory Network
provides comprehensive ecological data from
across the U.S.

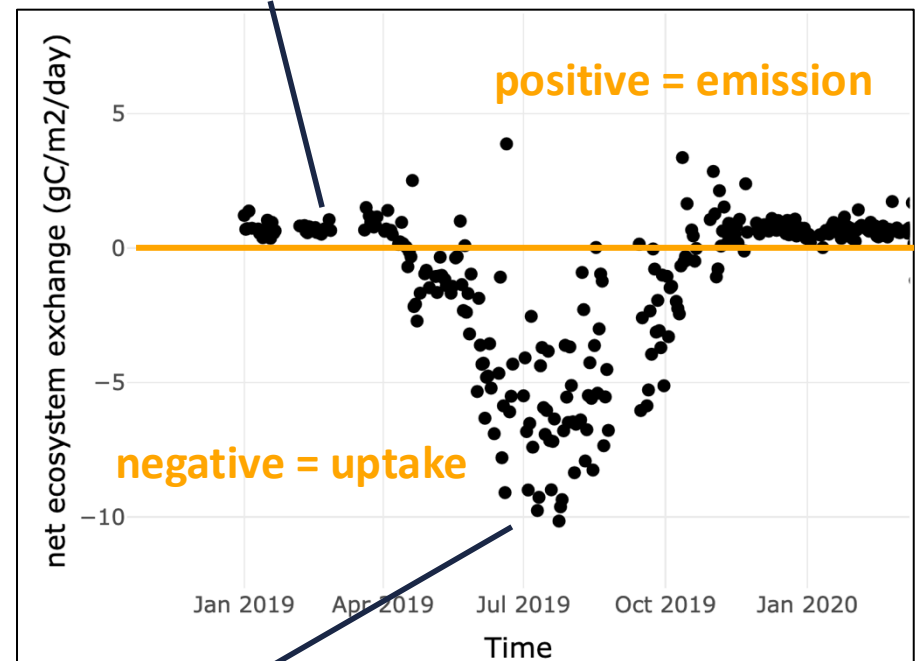


<https://www.neonscience.org/>

Target variable: net ecosystem exchange (NEE)

- **Net ecosystem exchange** is the net exchange of carbon dioxide between the air and an ecosystem
- NEE magnitude depends on the rates of photosynthesis and respiration of the ecosystem's vegetation
- If NEE is positive, the ecosystem is emitting carbon to the atmosphere; if it is negative, the ecosystem is taking up carbon

In winter, photosynthesis rates decrease so respiration leads to emission of carbon to the atmosphere from the ecosystem



In summer, high rates of photosynthesis lead to uptake of carbon from the atmosphere by the ecosystem

Climate implications of NEE

- NEE measures the exchange of **carbon dioxide**, a greenhouse gas
- Thus, NEE measures the degree to which an ecosystem contributes to or mitigates climate change
- Because NEE is affected by temperature and moisture, changes in climate can affect NEE, creating a feedback loop between ecosystems and climate



Image credit: U.S NEON

NEE is measured by high-frequency sensors deployed on towers which extend above the canopy of an ecosystem

Activity A: Visualize data from a selected environmental data case study, and fit and assess a time series model

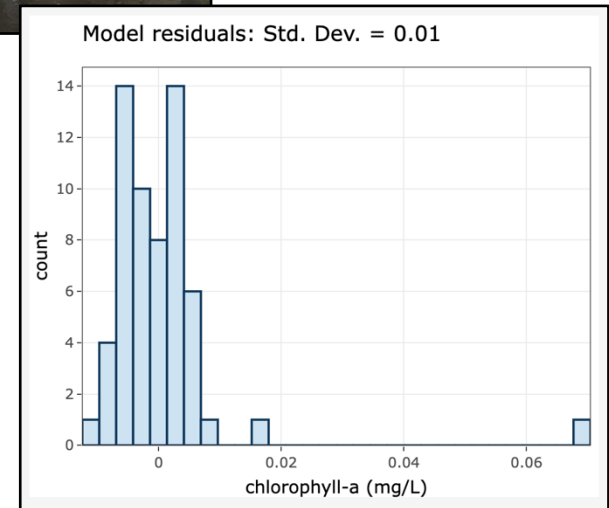
Objective 1: Select an environmental case study

Objective 2: Explore data

Objective 3: Learn about ARIMA time series model structure

Objective 4: Fit an ARIMA model to environmental case study data

Objective 5: Assess the ARIMA model fit



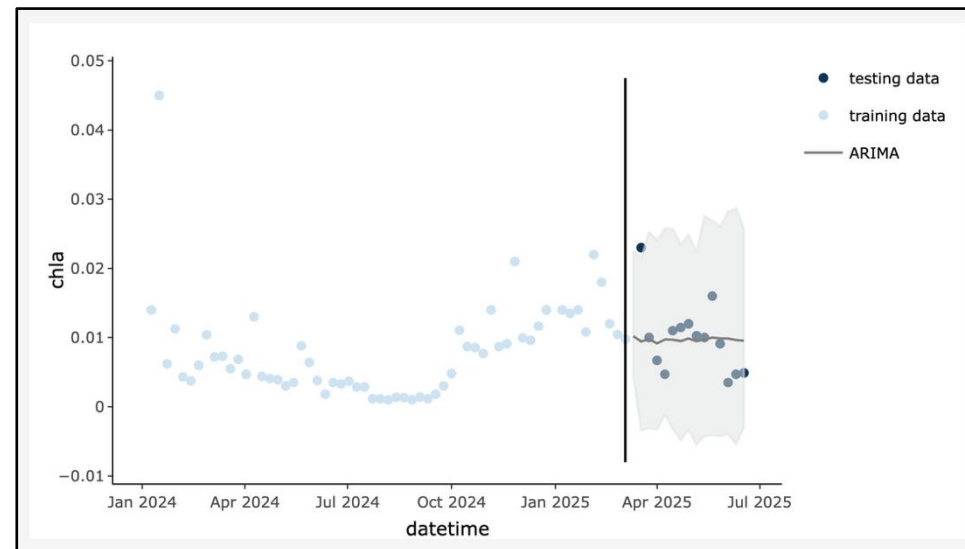
Activity B: Choose a new environmental dataset or upload your own dataset and fit and assess a time series model

Objective 6: Format and upload a new environmental data set of your choosing!

Objective 7: Fit an ARIMA model to your new environmental dataset

Objective 8: Assess the ARIMA model fit

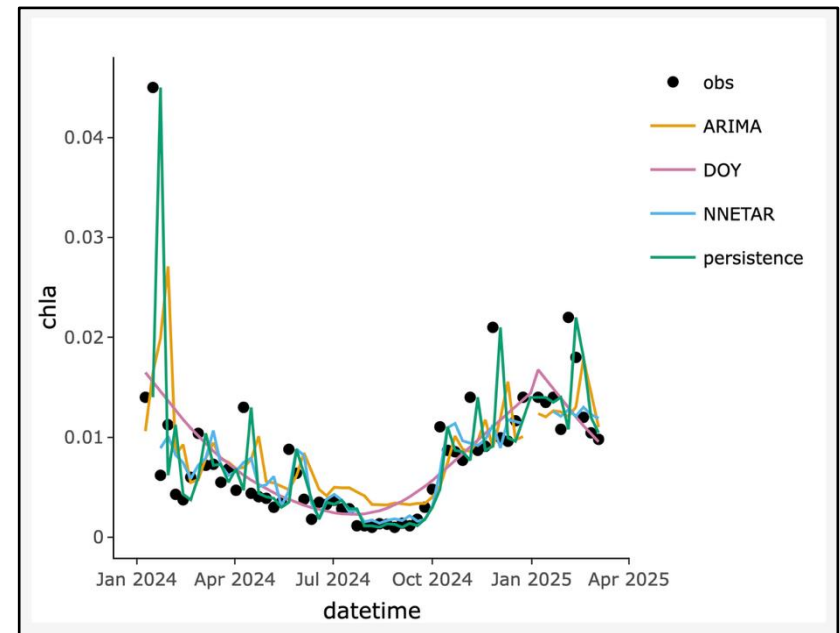
site_id	datetime	variable	observation
kent	2024-01-09	chla	0.01
kent	2024-01-09	amm	17.00
kent	2024-01-09	don	480.00
kent	2024-01-09	frp	5.00
kent	2024-01-09	oxygen	7.18



Activity C: Fit additional time series models to your environmental dataset and compare performance across models

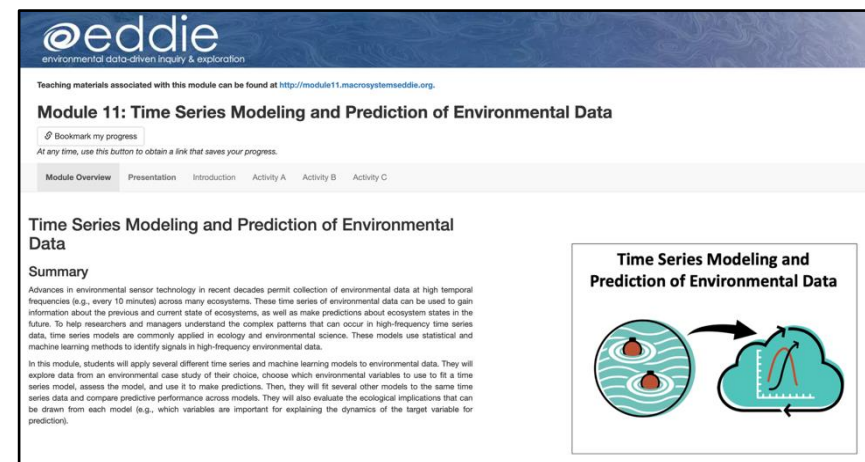
Objective 9: Fit additional models to the data you used in Activity B

Objective 10: Compare performance across models using out-of-sample predictions



R Shiny App

- You will complete module activities in an R Shiny app, which is an interactive website
- Be sure to complete the “Quick-start” guide to the module and watch the video that explains the interactive module features
- Questions are embedded in the app and you will answer these in a Word document that you download from the app



<http://macrosystemseddiedie.shinyapps.io/module11/>

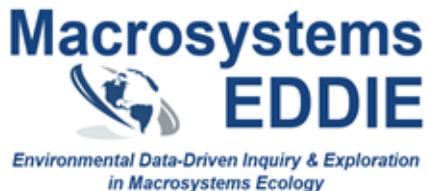
Thank you for participating!



Check out our other modules!

Find out more at:

MacrosystemsEDDIE.org



Swan-Canning Estuary Virtual Observatory



Note: All photos, graphs, and diagrams without credits were taken or developed by the module author team.